Practical : 1	To implement a goal-based agent using python that selects the correct action to achieve a specific goal.
Date : 05/08/26	

Aim: -

To implement a Goal-Based Agent using Python that selects the correct action to achieve a specific goal.

Algorithm:-

1. Start the program.
2. Import the required libraries.
3. Generate a random current position for the robot between 1 and 4.
4. Set the goal position as 5.
5. Compare the current position with the goal position.
6. If the current position is less than the goal position, move the robot forward.
7. Increase the current position by 1.
8. Repeat the process until the robot reaches the goal position.
9. Display the message that the goal has been reached successfully.
10. Stop the program.

Program:-

```
# ================================================================
# Practical No. 1
# Goal-Based Agent
# ================================================================

import random
import time
import os
import datetime

# Display Program Title
print("=" * 65)
print(" GOAL-BASED AGENT")
print("=" * 65)

# Display Date and Time
current = datetime.datetime.now()

print("\nDate :", current.strftime("%d-%m-%Y"))
print("Time :", current.strftime("%H:%M:%S"))

# Display Operating System
print("\nOperating System :", os.name)

# Generate Random Starting Position
current_position = random.randint(1, 4)

# Goal Position
goal_position = 5

print("\nRobot Current Position :", current_position)
print("Goal Position :", goal_position)

print("\nAI is Planning Path...")
time.sleep(2)

# Goal-Based Decision
while current_position < goal_position:
```

```
    print("Robot Moving Forward...")
    time.sleep(1)

    current_position += 1

    print("Current Position :", current_position)

# Goal Achieved
print("\nGoal Reached Successfully!")
print("Robot Arrived at Position", goal_position)

print("\nTask Completed Successfully.")
print("=" * 65)
```

Output:

```
C:\Users\Vishnu\OneDrive\AI_Practicals>python practical1_goal_based.py
=================================================================
 GOAL-BASED AGENT
=================================================================

Date : 23-08-2026
Time : 22:31:51

Operating System : nt

Robot Current Position : 1
Goal Position : 5

AI is Planning Path...
Robot Moving Forward...
Current Position : 2
Robot Moving Forward...
Current Position : 3
Robot Moving Forward...
Current Position : 4
Robot Moving Forward...
Current Position : 5

Goal Reached Successfully!
Robot Arrived at Position 5

Task Completed Successfully.
=================================================================
```

Conclusion:

Thus, the Goal-Based Agent was successfully implemented using Python. The agent compares the current position with the goal position and selects the appropriate action, moving forward until the specified goal is achieved**.**